How-To Library

Informatica

Migrating Mappings and Mapplets from a PowerCenter Repository to a Model Repository

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Abstract

Informatica supports the migration of mappings and mapplets created with the PowerCenter Client to a Model repository. This article explains how you can import objects from a PowerCenter repository into a Model repository. The article also outlines guidelines and restrictions to consider, and notes changes to objects that might occur during migration.

Supported Versions

- Informatica Big Data Management 9.6.1, 10.0, 10.1.x, 10.2.x
- Informatica Data Quality 9.6.1, 10.0, 10.1.x, 10.2.x
- Informatica Data Services 9.6.1, 10.0, 10.1.x, 10.2.x
- Informatica PowerCenter 9.1, 9.5.1, 9.6.1, 10.1.x, 10.2.x

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Overview

Certain use cases might require you to migrate mappings and mapplets from a PowerCenter repository to a Model repository.

For example, you might want to deploy Hadoop in your data warehouse environment. You might have mappings in PowerCenter that you can use for this purpose. You can import the mappings into the Model repository to take advantage of the big data processing speed available through the Developer tool.

You can use the Developer tool to import objects from a PowerCenter repository directly into a Model repository. You must be able to connect to the PowerCenter repository to use this functionality.

Due to differences between PowerCenter and the Model repository, you might not be able to import all dependent objects within a PowerCenter mapping or mapplet. The import process might also change some objects during migration.

What to Know Before You Migrate

PowerCenter objects contain properties that are not supported in the Developer tool and that cannot be processed by the Data Integration Service.

When an unsupported property is encountered during conversion, the import process might take one of the following actions:

- Ignore. Ignores the property and imports the object.

- Convert internally. Imports the object with the property but the Developer tool does not display the property.
- Fail import. The object import fails.

Review the guidelines and restrictions to avoid potential issues during migration and to understand what changes the migration process might make to objects.

General Conditions and Restrictions

The following conditions and restrictions apply to certain types of objects imported from PowerCenter:

Validation of sources or targets from PowerCenter release 9.1.0

When you import a source or target from PowerCenter release 9.1.0, the import process cannot verify if a connection type associated with the object is valid.

Invalid row delimiters

If the row delimiter for a flat file source is not valid, the import process changes it to the default value.

Mapping pipelines

An imported mapping can contain only one pipeline.

IBM DB2 source and target names

If the version of the PowerCenter repository is earlier than 9.5.0, an IBM DB2 source database name or an IBM DB2 target name must start with "DB2" to set the DB2 type.

Netezza or Teradata

Effective in Informatica 10.1, you can import Netezza and Teradata sources and targets.

When you import Netezza or Teradata sources, unsupported properties such as Foreign Key and Number of Sorted Ports are ignored.

When you import Netezza or Teradata targets, unsupported properties such as Foreign Key and Update Override are ignored.

Transformation Guidelines and Restrictions

The following conditions and restrictions apply to transformations imported with a mapping or mapplet into a Model repository.

Unsupported properties of certain PowerCenter transformations import with restrictions or are ignored by the import process. Some transformations are incompatible with the Model repository and fail to import.

Transformations that fail to import

The following transformation types fail to import:

- Data Masking
- External Procedure
- HTTP
- Identity Resolution
- Stored Procedure
- Transaction Control
- Unstructured Data
- XML Generator

- XML Parser

Aggregator transformations

An expression in an Aggregator transformation must contain 4,000 characters or fewer. If the value of the Data Cache Size or the Index Cache Size properties is not valid, the import process changes the value to Auto. The Transformation Scope property is ignored.

Java transformations

In a Java transformation, the ports must be input ports or output ports. The import fails if the Java transformation has both input ports and output ports.

Unsupported transformation properties such as Class Name and Function Identifier, in addition to certain output properties, are ignored.

Joiner transformations

If the value of the Data Cache Size or the Index Cache Size properties is not valid, the import process changes the value to Auto. The Null Ordering in Master, Null Ordering in Detail and Transformation Scope properties convert internally.

Lookup transformations

The database type for a Lookup transformation converts to ODBC during the import process.

The Lookup Sql Override property is converted to the Custom SQL Query property. Unsupported properties such as Recache if Stale and Synchronize Dynamic Cache are ignored.

Normalizer transformations

When you import a Normalizer transformation into the Developer tool, the Normalizer transformation imports with one input group and at least one output group.

The Developer tool might create multiple output groups based on links to the downstream transformation or targets in the mapping. For more information about the rules and guidelines about links from the multi-group transformation to the target, see the *Developer Transformation Guide*.

When you import a mapping that contains a reusable Normalizer transformation, the Developer tool imports the transformation as reusable. The Developer tool also replaces the reusable Normalizer transformation instances in the mapping with non-reusable transformation instances. The Developer tool generates new links from the non-reusable Normalizer transformation to the downstream transformations and target.

In PowerCenter, the Normalizer transformation has at least one generated key port. In the Developer tool, the Normalizer transformation does not contain a generated key port. When you import a Normalizer transformation from PowerCenter, the Developer tool ignores the generated key port. The Reset and Restart properties are also ignored.

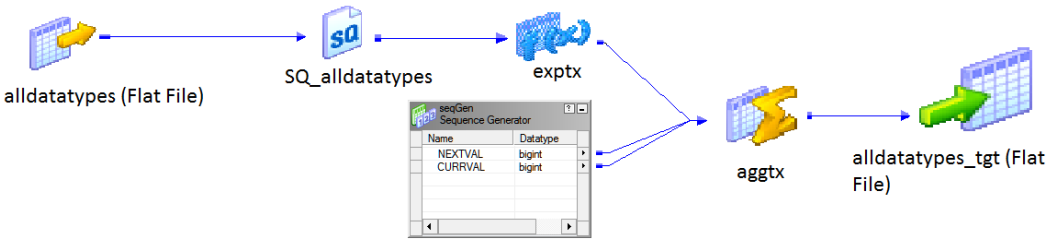
Rank transformations

An expression in a Rank transformation must contain 4,000 characters or fewer. If the value of the Data Cache Size or the Index Cache Size properties is not valid, the import process changes the value to Auto. The Transformation Scope property is ignored.

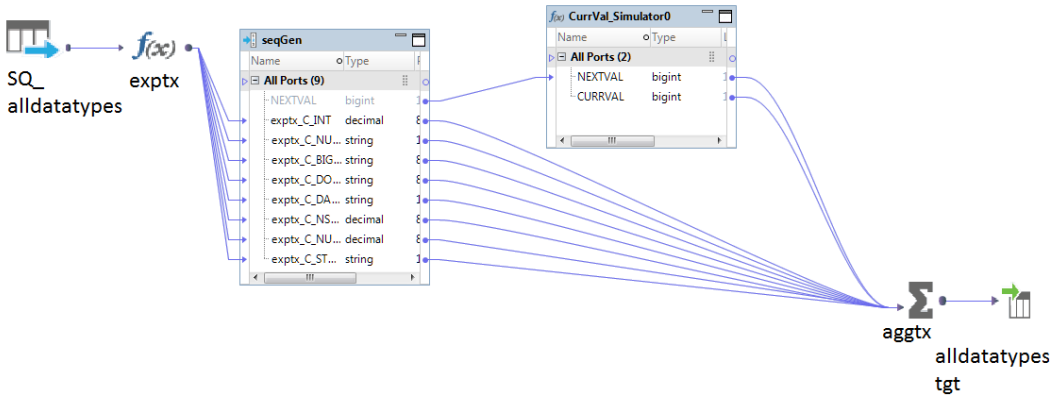
Sequence Generator transformations

In PowerCenter, the Sequence Generator transformation has two ports, CURRVAL and NEXTVAL. In the Developer tool, the Sequence Generator transformation has only one port, NEXTVAL. When you import a Sequence Generator transformation, the Developer tool creates an Expression transformation to set the values for the CURRVAL port and pass them to the downstream transformation.

The following image shows the PowerCenter mapping with a Sequence Generator transformation, Aggregator transformation, and an Expression transformation:



The following image shows the mapping imported into the Developer tool:



When you import the mapping, the Developer tool passes the NEXTVAL data from the Sequence Generator transformation to an Expression transformation to set the CURRVAL value.

The Developer tool imports a reusable Sequence Generator transformation in the mapping as a non-reusable transformation with a reusable sequence data object. When you import a reusable Sequence Generator transformation that is not part of the mapping, the Developer tool creates a reusable sequence data object.

The Current Value and Number of Cached Values properties are ignored.

Sorter transformations

The Transformation Scope property is ignored.

Source Qualifier transformations

A source object and a Source Qualifier transformation import completely as one data object. The Number of Sorted Ports property is ignored.

SQL transformations

The database type for an SQL transformation converts to ODBC during the import process.

The following import actions apply to SQL transformation properties:

- Connection Type fails to import if set to dynamic connection object or full dynamic connect information.
- Database Type fails to import if the value is set to Sybase, Informix, or Teradata.
- SQL Mode fails to import if set to script mode.
- Treat DB Connection Failure as Fatal is converted internally.

Union transformations

Unsupported transformation properties such as Class Name, Function Identifier and Transformation Scope, in addition to certain output properties, are ignored.

Importing Objects

Use the Developer tool to import objects from a PowerCenter repository into a Model repository.

1. Select **File** > **Import**.

   The **Import** dialog box appears.
2. Select **Informatica** > **PowerCenter**.
3. Click **Next**.

   The **Import from PowerCenter** dialog box appears.
4. Specify the connection parameters for the PowerCenter repository.
5. Click **Test Connection**.

   The Developer tool tests the connection to the PowerCenter repository.
6. If the connection to PowerCenter repository is successful click **OK**. Click **Next**.

   The Developer tool displays the folders in the PowerCenter repository and prompts you to select the objects to import.
7. Select one or more objects to import.
8. Click **Next**.
9. Select a target location for the import objects in the Model repository.
10. Select a conflict resolution option for object name conflict. Choose to rename, replace, or reuse the object in the target Model repository.
    - To rename the PowerCenter repository object with the default naming convention, and then import it to the Model repository, select the **Rename object in target** option. The default conflict resolution is to rename the object.
    - To replace the Model repository object with the PowerCenter repository object, select the **Replace object in target** option.
    - To reuse the object in the Model repository in the mapping instead of importing the PowerCenter object, select the **Reuse object in target** option.
11. Click **Next**.

    The Developer tool shows the PowerCenter objects, and the dependent objects.
12. Click **Ignore Overridden Properties** to ignore override properties for reusable PowerCenter sources, targets, and transformations. By default, the process preserves override properties.
13. If you import an IBM DB2 object, select the DB2 object type. You can select one of the following object types: LOW, z/OS, i5/OS.
14. Click **Next**.
15. Specify the Model repository connection details to the PowerCenter repository objects.
16. The **Choose Connection** dialog box appears. Select a connection and click **OK**.
17. Click **Next**.

    The Developer tool generates an import summary and lists the PowerCenter objects and the dependent objects to be imported.
18. Click **Conversion Check** to verify if objects can be imported as valid Model repository objects.

The Developer tool shows a conversion check summary report with results of the conversion check.

19. Click **OK**. Click **Finish**.

    The Developer tool shows progress information during the import. The Developer tool imports the PowerCenter objects and the dependent objects to the Model repository, and generates a final import summary report.

20. Click **Save** and specify a file name to save the import summary if there are conversion errors.

Migration Results

You might need to update objects after you import them into the PowerCenter repository. You might also notice differences between the Model repository and PowerCenter objects.

Changes to Data Types

Some PowerCenter datat ypes are not valid in the Model repository. When you import PowerCenter objects with data types that are not valid, the import process converts them to valid, comparable data types in the Model repository.

The following table lists the PowerCenter repository data types that convert to corresponding Model repository data type in the import process:

PowerCenter Repository Data Type	Model Repository Data Type
Real	Double
Small Int	Integer
Nstring	String
Ntext	Text

Author

Dan Hynes